

Sentinel

A Fully Procedural Game Engine
and Open-World Military Shooter

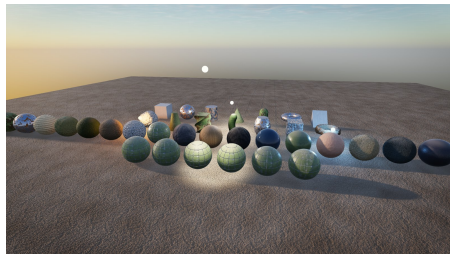
Nothing is loaded from a model, image or sound file:
everything is generated from code and a seed.

Md. Shifat Hasan Roll 2107067

Instructors: Md Tajmilur Rahman, Md. Mubtashim Abrar Nihal
4th Year 1st Term | October 7, 2026

Outline

1. Problem, objectives, architecture
2. Procedural shapes and textures
3. The rendering pipeline
4. World, physics and gameplay
5. Results and verification
6. Challenge, limits, future work
7. Video tour of the project



Every material is a code-generated recipe.

1. Problem, objectives and architecture

Can a complete, playable 3D game be made by code alone on a student laptop?

Two hard limits

- No model, image or precomputed data file
- macOS OpenGL 4.1: no compute shaders

Objectives

- Layered engine, downward dependencies
- All content from code and seeds
- Modern renderer, playable game
- Verified: tests, 16.7 ms budget

Game: mission, AI, vehicles, sound

Render: deferred, lights, sky, post

World: collision, terrain

Procedural: noise, textures

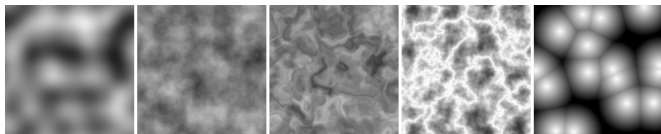
GPU: shaders, buffers

Platform: window, input



A lower layer never knows a higher one.

2. Procedural shapes and textures



gradient, fractal, warped, ridged and cellular noise

Fractal noise (tiles exactly)

$$F(\mathbf{x}) = \frac{\sum_{i=0}^{n-1} 2^{-i} v(2^i \mathbf{x})}{\sum_{i=0}^{n-1} 2^{-i}}$$

Vertex push

$$\mathbf{p}' = \mathbf{p} + A v(f\mathbf{p}) \hat{n}, \quad |\mathbf{p}' - \mathbf{p}| \leq A$$

- 18 recipes $\rightarrow$ 22 materials, baked once at 1024^2
- **Scharr** filter turns height into bumps; crevices darken
- Objects = tables of primitives + noise deformers
- Soldiers: skeletons with two-bone IK legs

3. The rendering pipeline



Deferred shading: store each pixel's colour, normal, shininess and depth first, then light it once.

Cook–Torrance reflection

$$f = \frac{(1-F_l)(1-F_v)c_d}{\pi} + DVF$$

$$D = \frac{\alpha^2}{\pi((\mathbf{n} \cdot \mathbf{h})^2(\alpha^2 - 1) + 1)^2}, F = F_0 + (1 - F_0)(1 - \mathbf{v} \cdot \mathbf{h})^5$$

Fog and tone map

$$C = TC_s + (1 - T)C_h, T = e^{-kd}$$

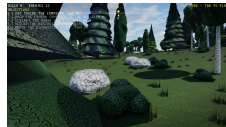
$$C_{\text{out}} = \frac{x(2.51x+0.03)}{x(2.43x+0.59)+0.14}$$



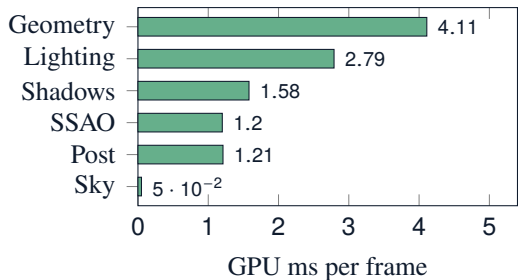
Dusk: sun, sky, haze, shadows

4. World, physics and gameplay

- **Sky:** sun height from astronomy; scattering; wind-driven clouds
$$\mathbf{P}_{xz} = \frac{H}{d_y}(d_x, d_z) + \mathbf{w}t$$
- **Terrain:** warped ridged noise, streamed in 64 m chunks
- **Trees:** 4 species, swaying in the wind
- **Vehicles:** jeep, truck, carrier, tank, helicopter; $\dot{\psi} = \frac{v \tan \delta}{L}$
- **Ladders:** 4-state grip machine
- **Mission:** hack, destroy, rescue, extract; stealth AI; synthesised sound



5. Results and verification



10.9 ms of the 16.7 ms budget at 1080p

Unit tests (13 packages)	281
TextureCheck (exact tiling)	8 318
RenderCheck	318 567
GpuCheck	30
Playthroughs, 5/5 objectives	2



6. Challenge solved, limits and future work

Bug: a bright vertical line at night. The fog colour sampled the *whole* sky, so one horizon star painted a streak on every surface in its column. Fix: fog uses the smooth atmosphere only.



before



after

Limitations

Blocky primitive models; no true global illumination; lumpy foliage; tuned by scripts and screenshots.

Future work

Leaf cards, screen-space GI, point-light shadows, multiplayer, a compute-shader port.

Take-away

Most “art” is a few simple ideas, applied consistently and tested.

7. Video tour of the project

References

- [1] K. Perlin, “Improving noise,” *ACM TOG*, 21(3), 2002.
- [2] S. Worley, “A cellular texture basis function,” *SIGGRAPH*, 1996.
- [3] R. Cook, K. Torrance, “A reflectance model for computer graphics,” *ACM TOG*, 1(1), 1982.
- [4] B. Walter et al., “Microfacet models for refraction through rough surfaces,” *EGSR*, 2007.
- [5] B. Karis, “Real shading in Unreal Engine 4,” *SIGGRAPH Course*, 2013.
- [6] S. Hillaire, “A scalable and production ready sky and atmosphere rendering technique,” *CGF*, 39(4), 2020.
- [7] R. Dimitrov, “Cascaded shadow maps,” *NVIDIA*, 2007.
- [8] M. Oren, S. Nayar, “Generalization of Lambert’s reflectance model,” *SIGGRAPH*, 1994.
- [9] K. Narkowicz, “ACES filmic tone mapping curve,” 2016.
- [10] J. Jimenez, “Next generation post processing in Call of Duty: Advanced Warfare,” *SIGGRAPH*, 2014.
- [11] T. Lottes, “FXAA,” *NVIDIA*, 2009.
- [12] M. Segal, K. Akeley, *The OpenGL Graphics System, v4.1*, Khronos, 2010.
- [13] I. Quilez, “Domain warping,” *iquilezles.org*.
- [14] S. Hargreaves, M. Harris, “Deferred shading,” *NVIDIA*, GDC 2004.
- [15] H. Scharr, “Optimal operators in digital image processing,” Ph.D., Heidelberg, 2000.
- [16] T. Akenine-Möller et al., *Real-Time Rendering*, 4th ed., CRC Press, 2018.

Thank you Questions are welcome · Md. Shifat Hasan, Roll 2107067